TAMID QUANT — STRATEGY ONE PAGER

Bitcoin Digital Options: Learned Tail Model (#7)

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The Idea

Kalshi's hourly "Bitcoin above $K at 5pm ET" contracts overprice longshots in the last hours before expiry. The tail strategy sells overpriced low-tail YES and buys underpriced high-tail YES, sizing each day so the worst BTC settlement loses at most 3% of bankroll. Model #7 replaces the base's average YES rate per price range with a logistic regression, refit weekly on past days only, that predicts each contract's outcome from its market price, a volatility model's probability and the horizon. On the holdout (Jul 1 to Nov 11, 2026) it had a Sharpe ratio of 2.34 against 1.07 for the base and made +9.6%. Over all the data it is weaker (Sharpe 0.78, +$108): it lost $47 before 2026, before its edge appeared.

Mathematical Framework

Core Model: *Learned probability of YES, refit every 7 days on all prior tail quotes:*
$f = \sigma(\beta_0 + \beta_1 \cdot \text{logit}(\text{price}) + \beta_2 \cdot \text{logit}(p_{model}) + \beta_3 \cdot 1[H = 3])$, L2-penalized. Final weights: $\beta_1 = 0.96$, $\beta_2 = 0.28$, $\beta_3 \approx 0$. Because $\beta_1 + \beta_2 = 1.24 > 1$, longshots come out rarer than the market says: a 2c contract with the model at 2% gets f ≈ 0.7%.

Signal Construction: *Edge = f vs. the sizing price, less fee:*
The model probability p_{model} comes from an EWMA volatility forecast with filtered-historical-simulation tails. Sizing price = quote ± 2c slippage, clipped to 1c/99c; Kalshi fee $0.07 \cdot P \cdot (1 - P)$.

Entry / Exit Logic: *Same as the base: decide 6h and 3h before expiry, fill one hour later:*
Trade mid < 10c or > 90c, spread ≤ 6c, edge ≥ 0.5c, at least 100 prior observations in the price range. Hold to settlement.

Position Sizing: *Same linear program as the base:*
maximize $\Sigma\, q_i \cdot \text{edge}_i$ so that every BTC settlement scenario loses at most 3% of bankroll, $0 \le q_i \le 50$.

Selection: *Ten numbered models, tuned Jan to Jun 2026, scored once on Jul 1 to Nov 11:*
#7 and #5 (model prior) beat the base in both periods. Over all data #5 is steadier (Sharpe 1.17, drawdown −$31.54).

Key Risks

- Not significant: Sharpe minus base = +1.27 on the holdout (95% CI −2.9 to +5.2) and +0.31 over all data (95% CI −1.6 to +2.0)
- Uneven record: lost $47 before 2026 (3 tail days), fell to $931.75 in Apr 2026, max drawdown −$84.15 (base −$54.54, #5 −$31.54)
- Tail days remain: 11 days lost more than $15. The average winning trade makes +$1.41 and the average loser loses −$23.72
- The cap is breached more often: on 77 days the realized worst case was past 3% (worst 3.17%), against 24 for the base. Fewer trades sit at the 1c/99c clip, so the one-hour fill lag matters more
- A fitted formula can be confidently wrong when the market changes, and picking the best of 10 models adds luck

At a Glance

Asset Class:	*Crypto / Prediction Markets*
Strategy Type:	*Longshot Bias / Learned Calibration*
Direction:	*Short Low-Tail YES, Long High-Tail YES*
Holding Period:	*3-6 Hours (Hold to Expiry)*
Universe:	*Kalshi KXBTCD Tail Strikes*

Backtest Results

Period	#7 Sharpe	Base Sharpe	#7 profit
Before 2026	−0.97	n/a	−$47
Tuning	1.51	0.29	+$59
Holdout	2.34	1.07	+$96
All data	0.78	0.47	+$108

Tuning Jan-Jun 2026, holdout Jul 1-Nov 11 2026, all data Dec 23 2024-Nov 11 2026. Annualized Sharpe ratio; base made 2 trades before 2026. $1,000 to $1,107.72 over 688 days; 329 trades on 237 days; 3% worst-case cap per day.

Data Requirements

- Kalshi KXBTCD hourly bid, ask and settlement; Coinbase BTC-USD hourly candles - public APIs
- 815 expiries (Jul 2024 to Nov 2026), 36,093 quotes at 6h and 3h; the first year only trains the model

Key Parameters

- Logistic refit every 7 days; L2 = 1; f clipped to 0.1-99.9%
- Worst-case cap 3% of bankroll per expiry day
- Tails mid < 10c or > 90c at 6h and 3h; min edge 0.5c; slippage 2c; max 50 contracts per order

Open Questions

- Does it hold on data after Nov 11, 2026 (a true second holdout)?
- Was the pre-2026 loss from too little training data? Would a later start fix it?
- Does adding recent volatility or the time of day improve the fit?
- Is there depth to trade 30+ contracts at these prices?